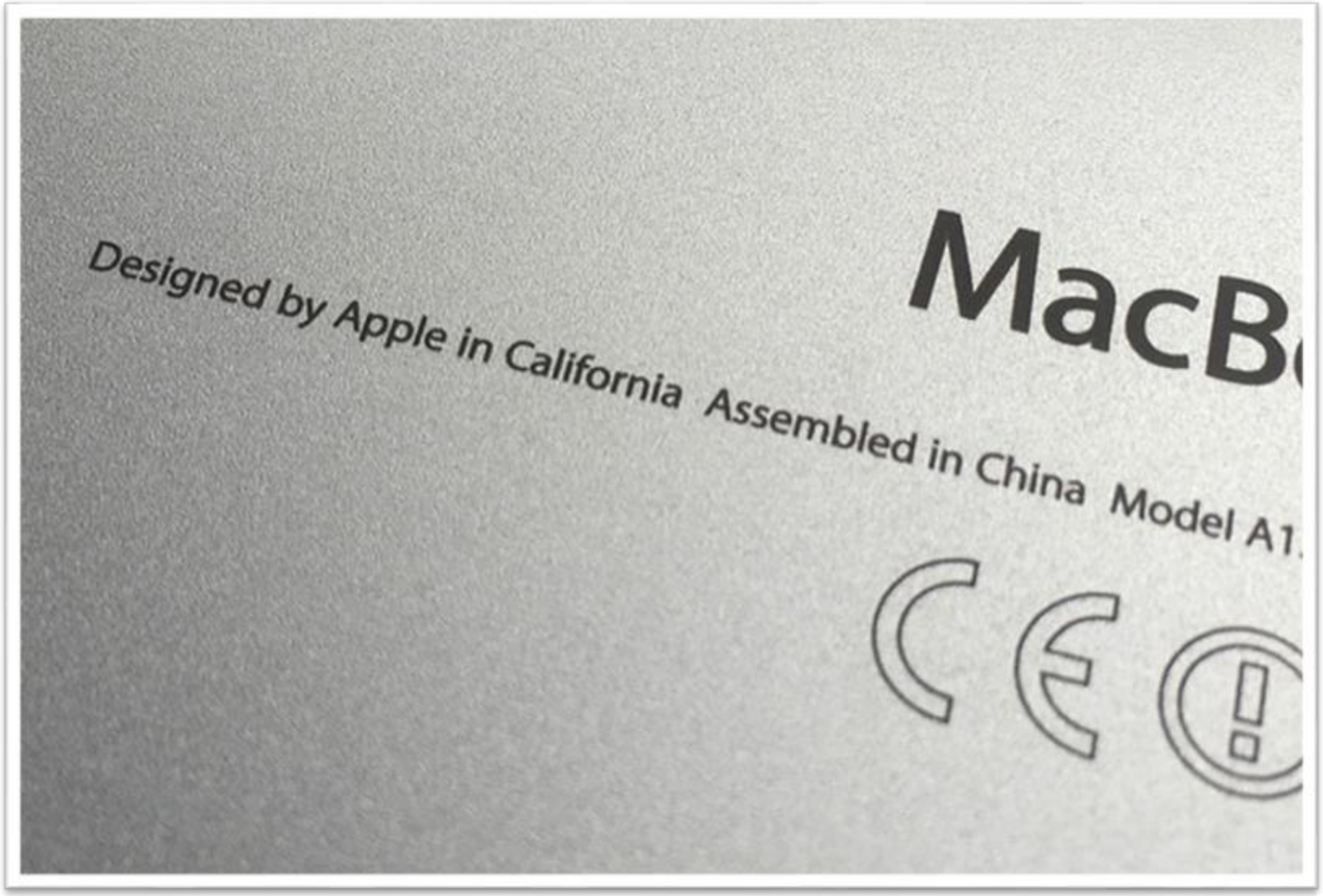


Lecture: Old Trade Theory

Attend your tutorial next week

- There will be a quiz, but does *not* count towards your mark. Its purpose is simply to familiarise you with the types of questions you can expect in the final exam.
- A combination of MCQs and short questions
- It will cover the first two lectures.

Lecture plan:

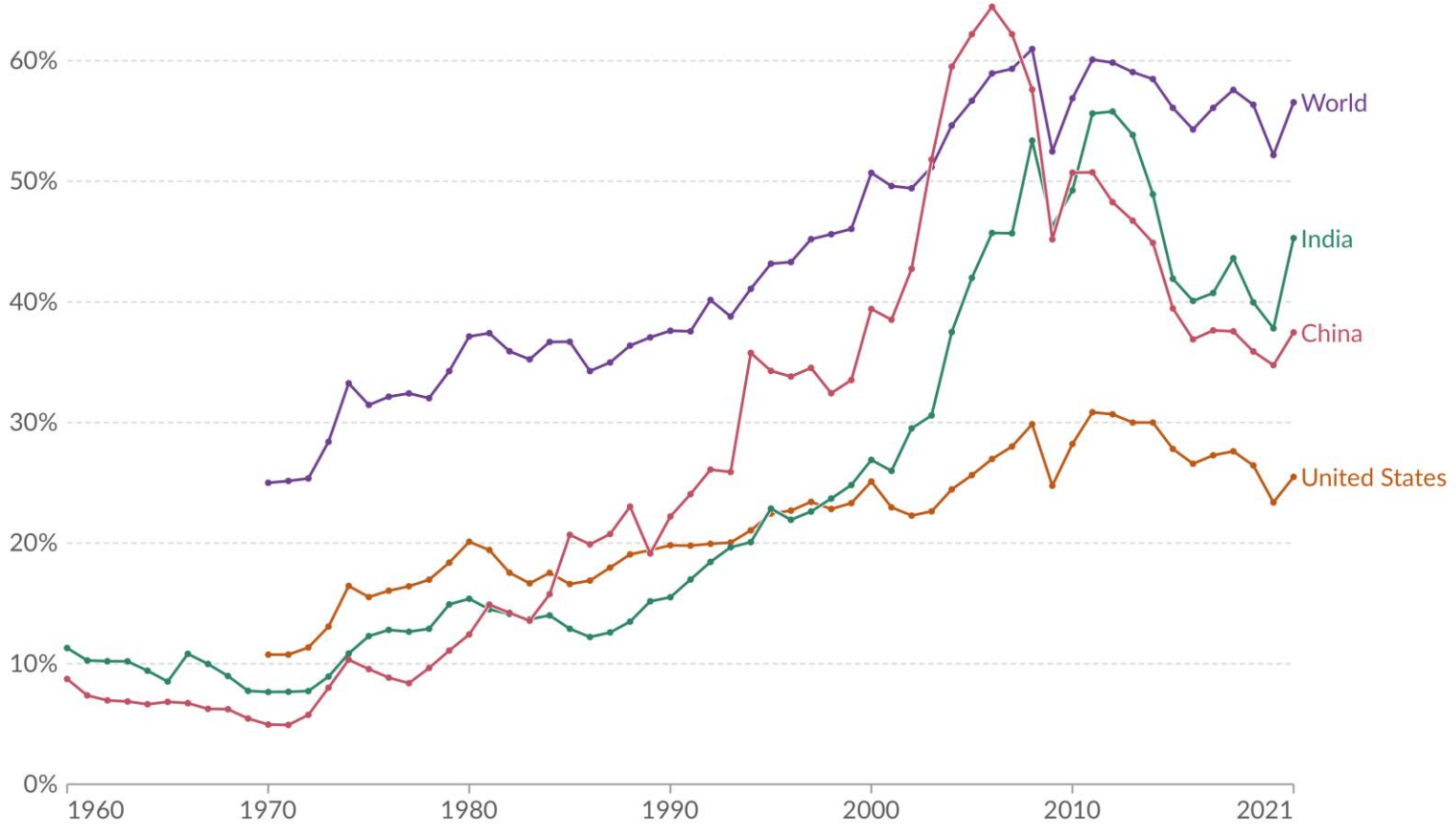
- Types of trade
- Theory of absolute advantage
- Theory of comparative advantage
- Shortcomings of these theories

Lecture plan:

- **Types of trade**
- Theory of absolute advantage
- Theory of comparative advantage
- Shortcomings of these theories

Trade as a share of GDP, 1960 to 2021

Sum of exports and imports of goods and services, divided by gross domestic product, expressed as a percentage. This is also known as the "trade openness index".



Why?

Data source: World Bank and OECD

OurWorldInData.org/trade-and-globalization | CC BY

Join at menti.com | use code 2542 0188

Wat have been the main drivers of trade growth since 1980?

0 responses

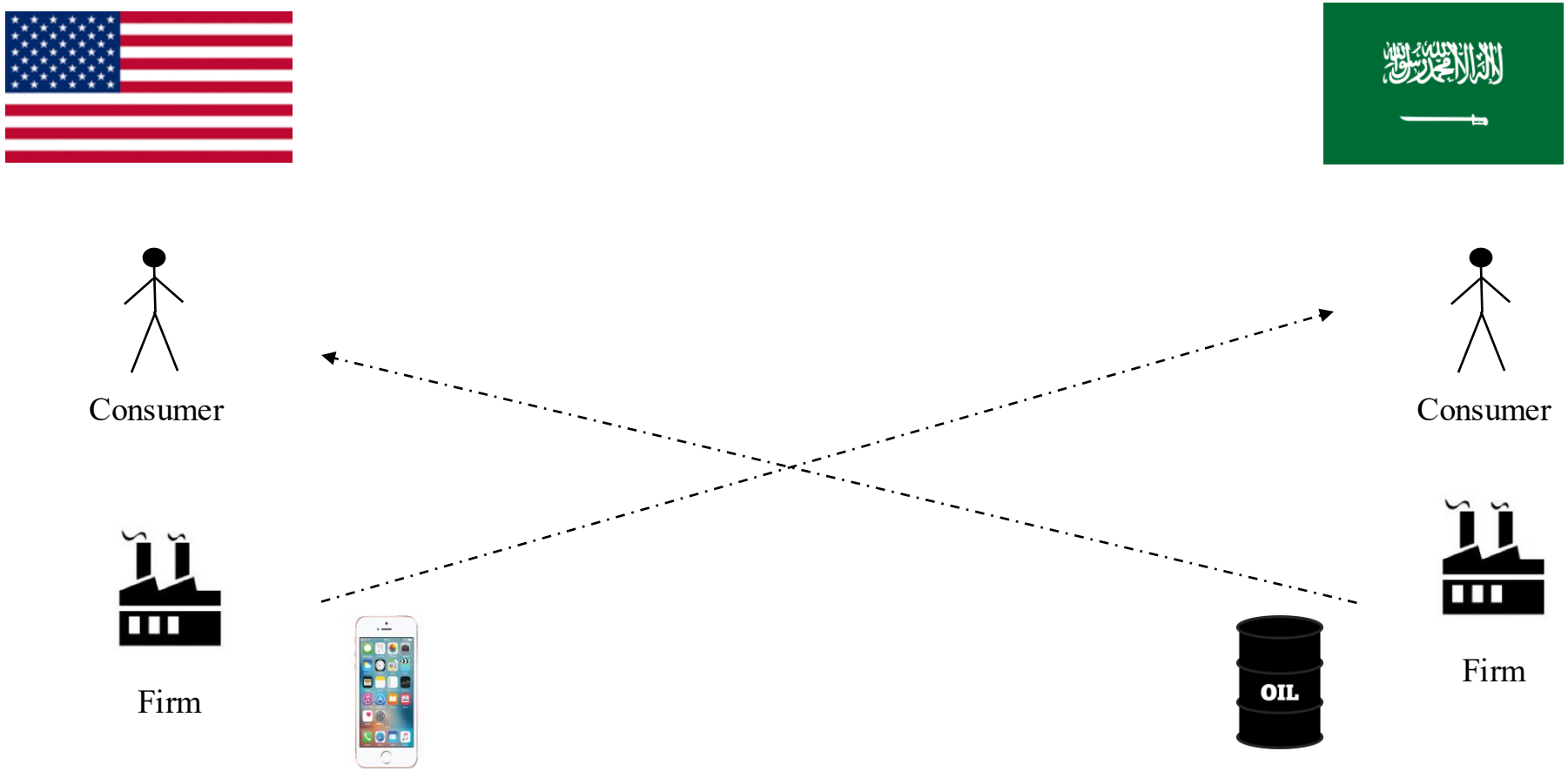
Responding is off

Press C to allow new responses

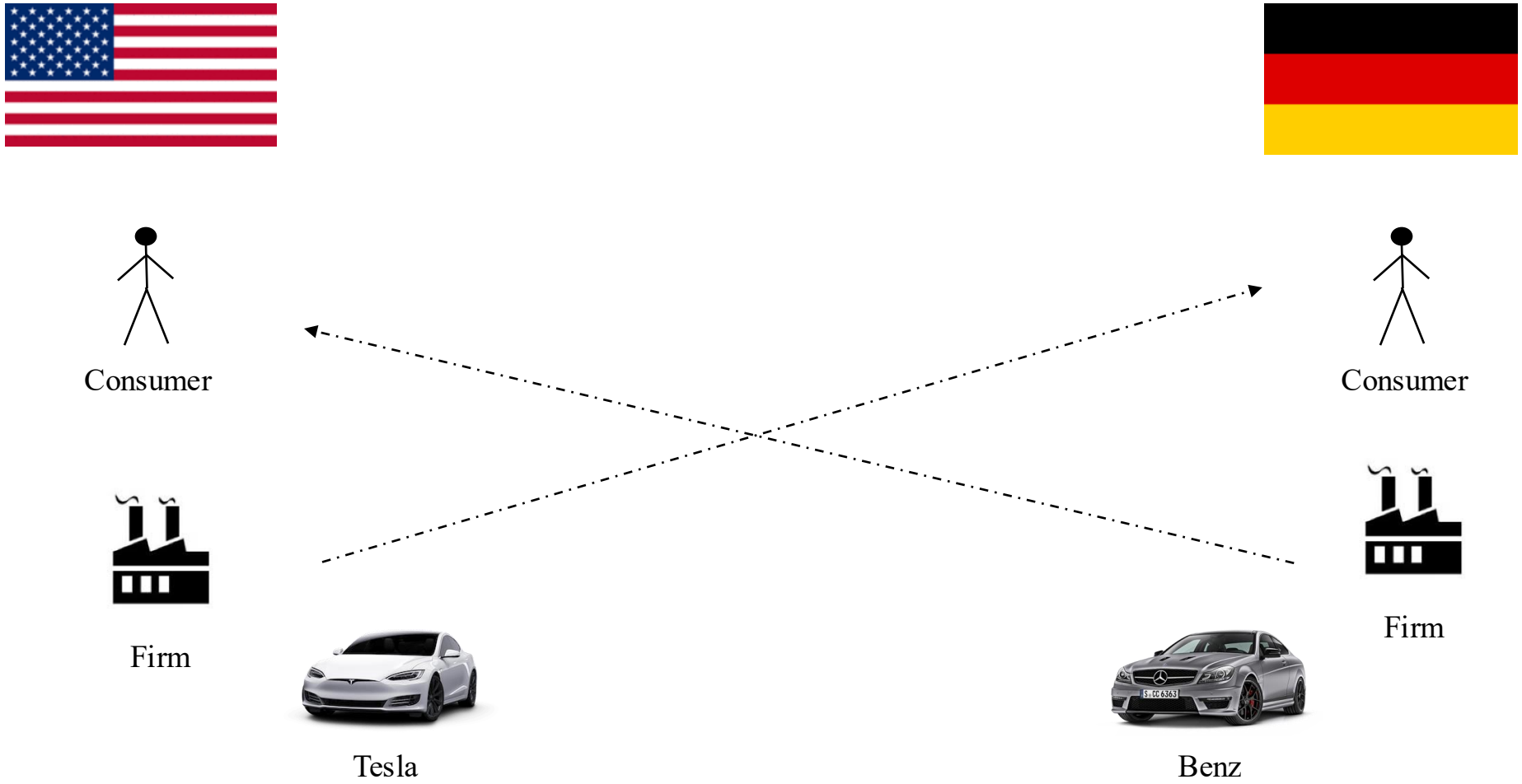
Table 4.1. Types of trade

Type of trade	Phrase	Meaning	
Inter-industry	Either/or	*Either* imports *or* exports in a given sector of the economy	North-South
Horizontal intra-industry	Both/and/same	*Both* imports *and* exports in a given sector of the economy and at the *same* stage of processing	North-North South-South
Vertical intra-industry (Outsourcing by firms)	Both/and/different	*Both* imports *and* exports in a given sector of the economy and at *different* stages of processing	North-South

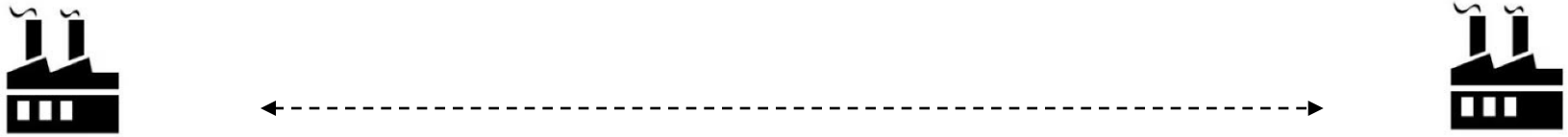
Trade in dissimilar goods (inter-industry)



Trade in similar goods (intra-industry)



Outsourcing: Trade in dissimilar goods/tasks in the same sector (intra-industry)



Apple

Foxconn

iPhone components

Assembly

Table 4.1. Types of trade

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Why do countries engage in these sort of trades?

Lecture plan:

- Types of trade
- **Theory of absolute advantage**
- Theory of comparative advantage
- Shortcomings of these theories

	Japan	Vietnam
	$100	$200
	$5	$2

Partial equilibrium
(analyzing each market on its own)

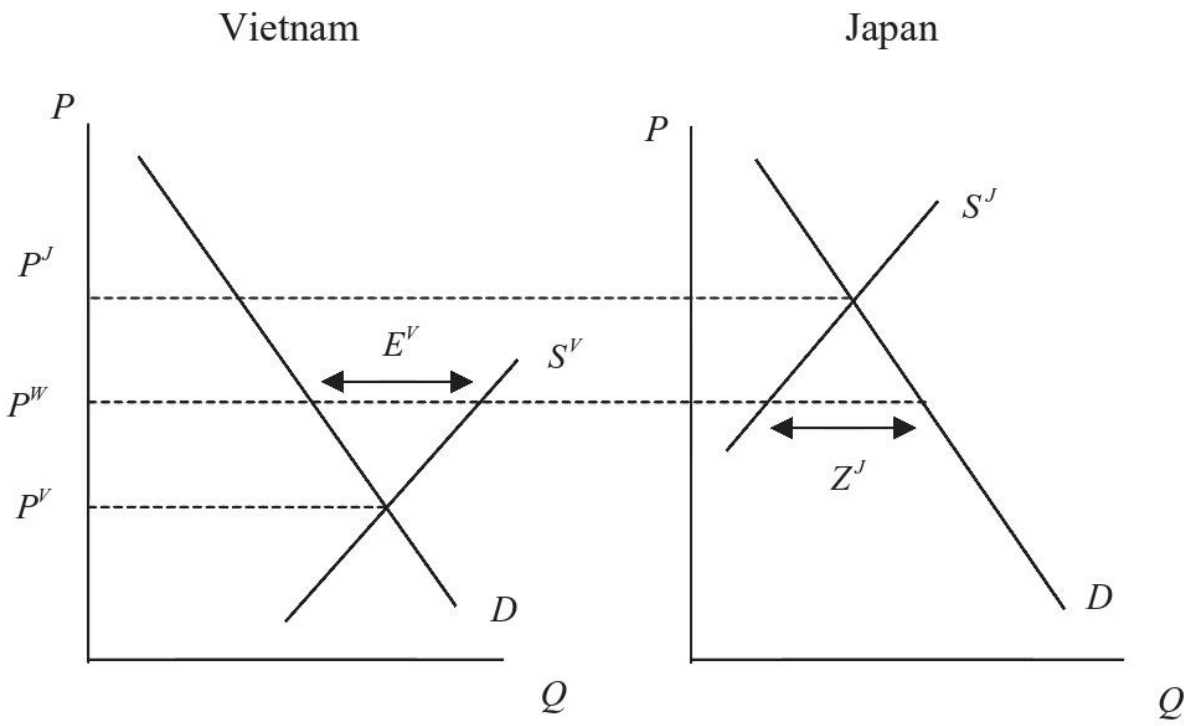


Figure 2.4. Trade in the Rice Market

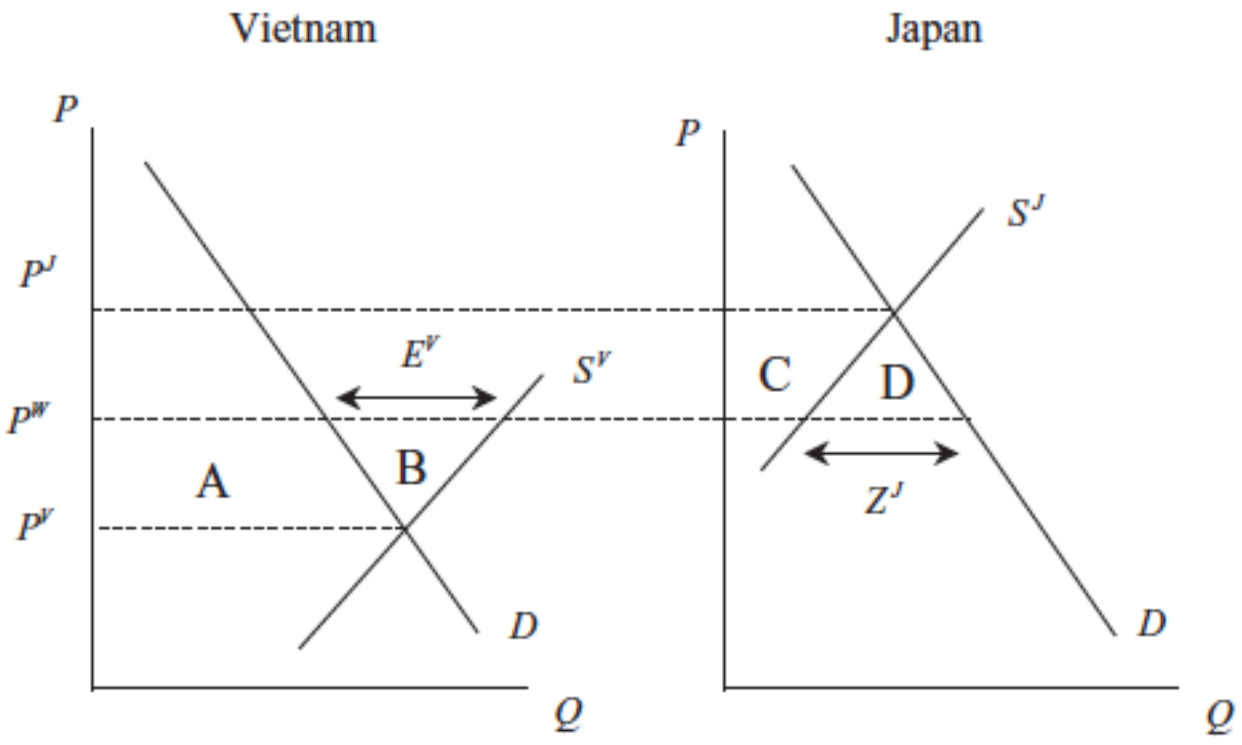


Figure 2.6. Gains from Trade in the Rice Market

Who are the winners & losers of trade within Vietnam?

General equilibrium
(Analyzing all markets at the same time)

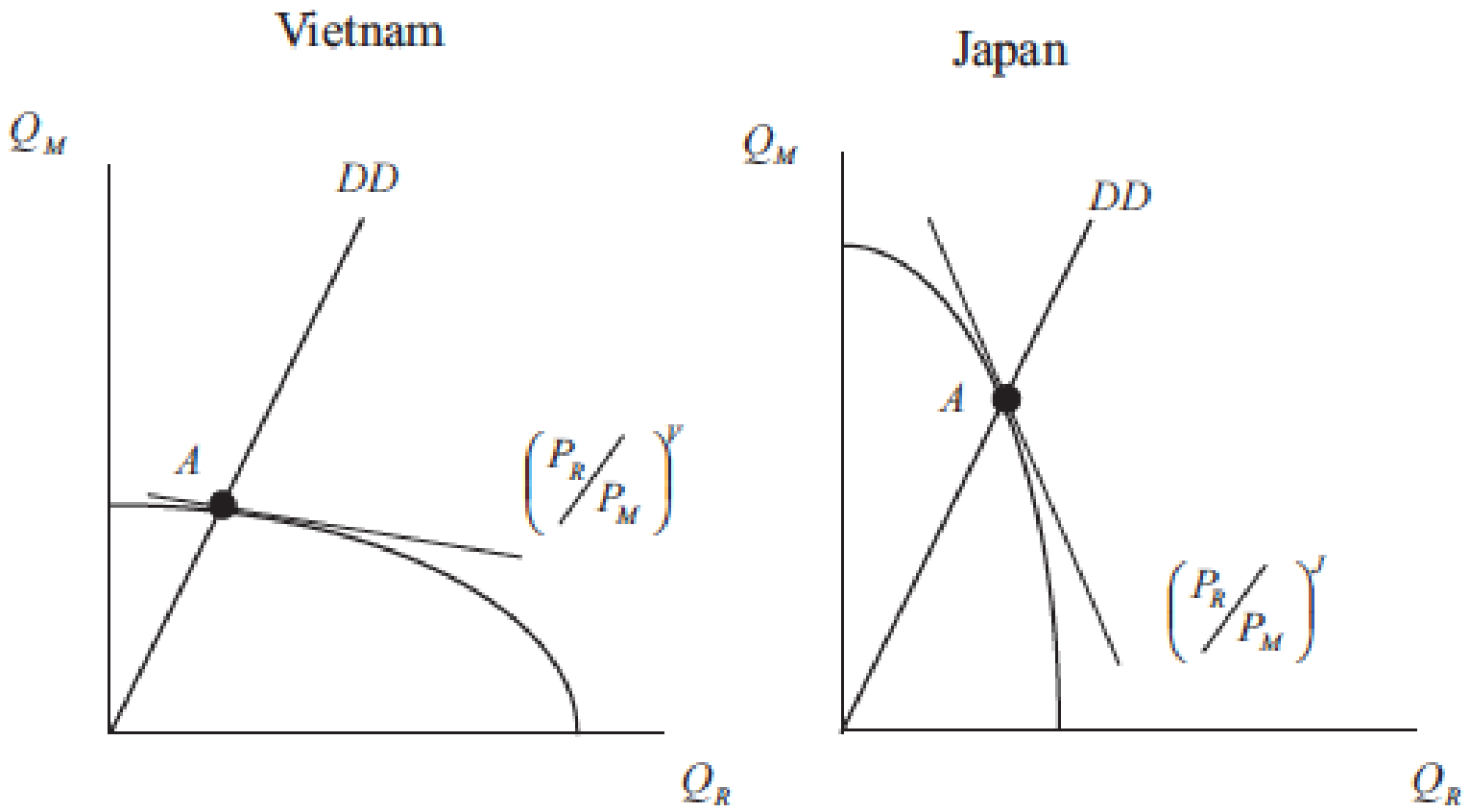
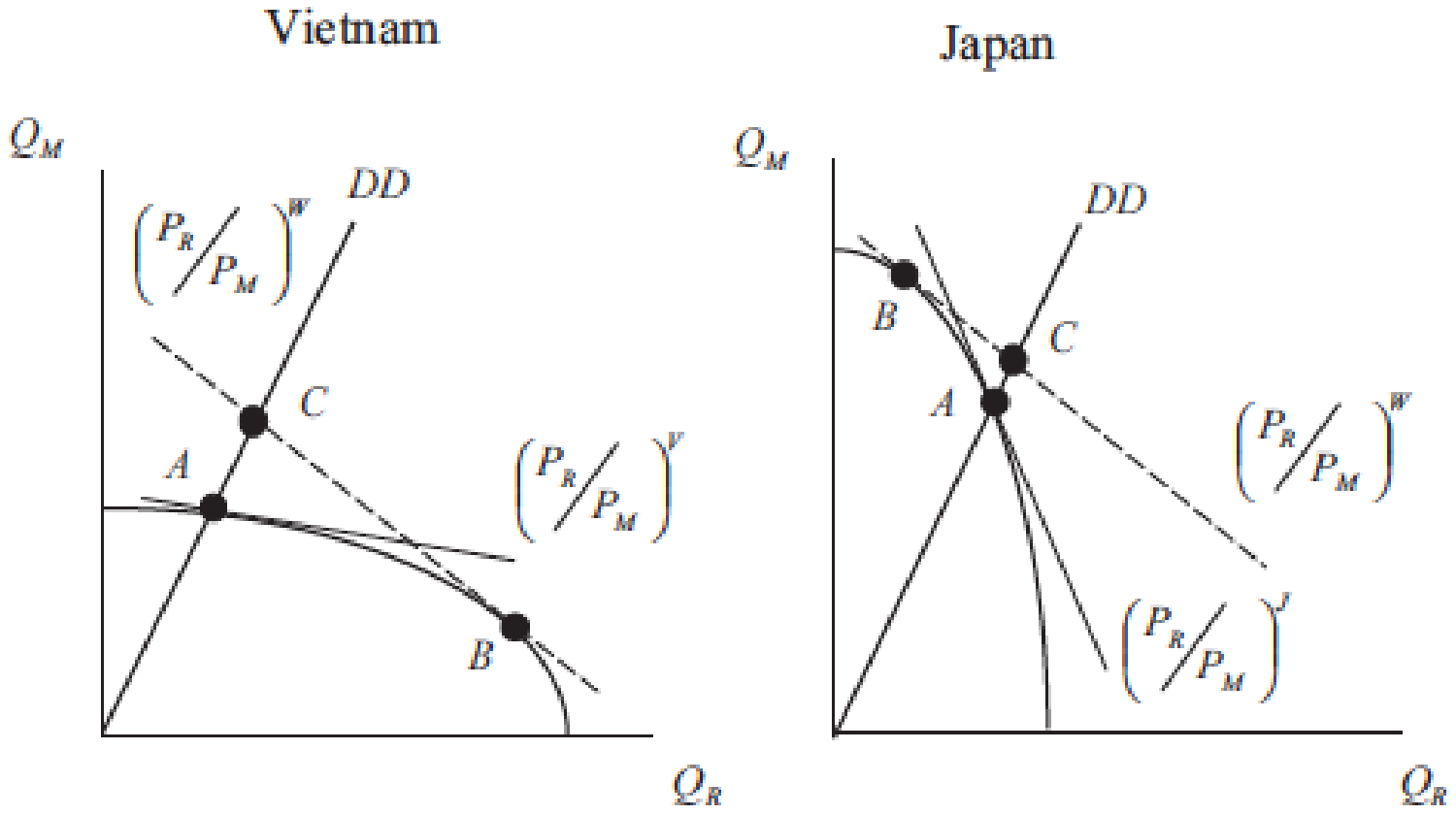
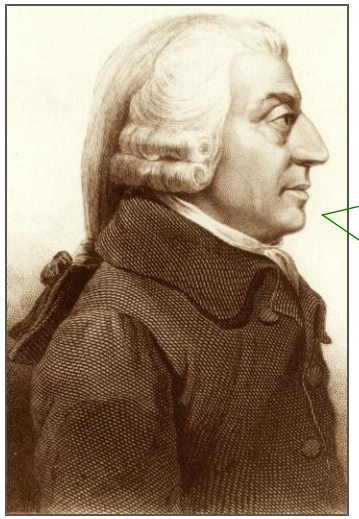


Figure 3.3. Relative Prices in Vietnam and Japan under Autarky



By specialisation and trade, each country can consume more

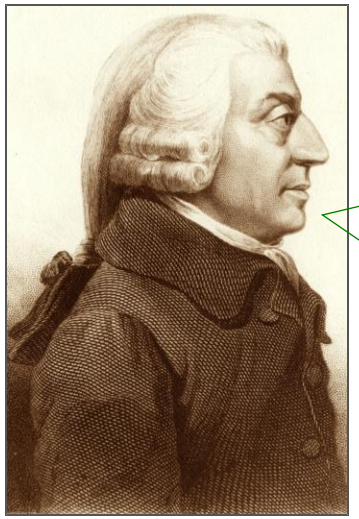
Theory of absolute advantage



Adam Smith
(1723-1790)

Free trade enables countries (or firms) to *specialise* in production of goods in which they have an absolute *cost/productivity advantage* (i.e. price advantage) and import goods (or outsource tasks) in which they have a cost disadvantage relative to other countries.

Theory of absolute advantage



Adam Smith
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Free trade enables countries (or firms) to *specialise* in production of goods in which they have an absolute *cost/productivity advantage* (i.e. price advantage) and import goods (or outsource tasks) in which they have a cost disadvantage relative to other countries. **Each country export goods at a higher price than home prices so that they can import goods at lower prices than home prices.** Specialisation enables countries to *consume* more at the aggregate level!

Assumptions of the Smith's model:

Difference in production costs (Wage, production knowledge) → Absolute Price difference ← Differences in demand

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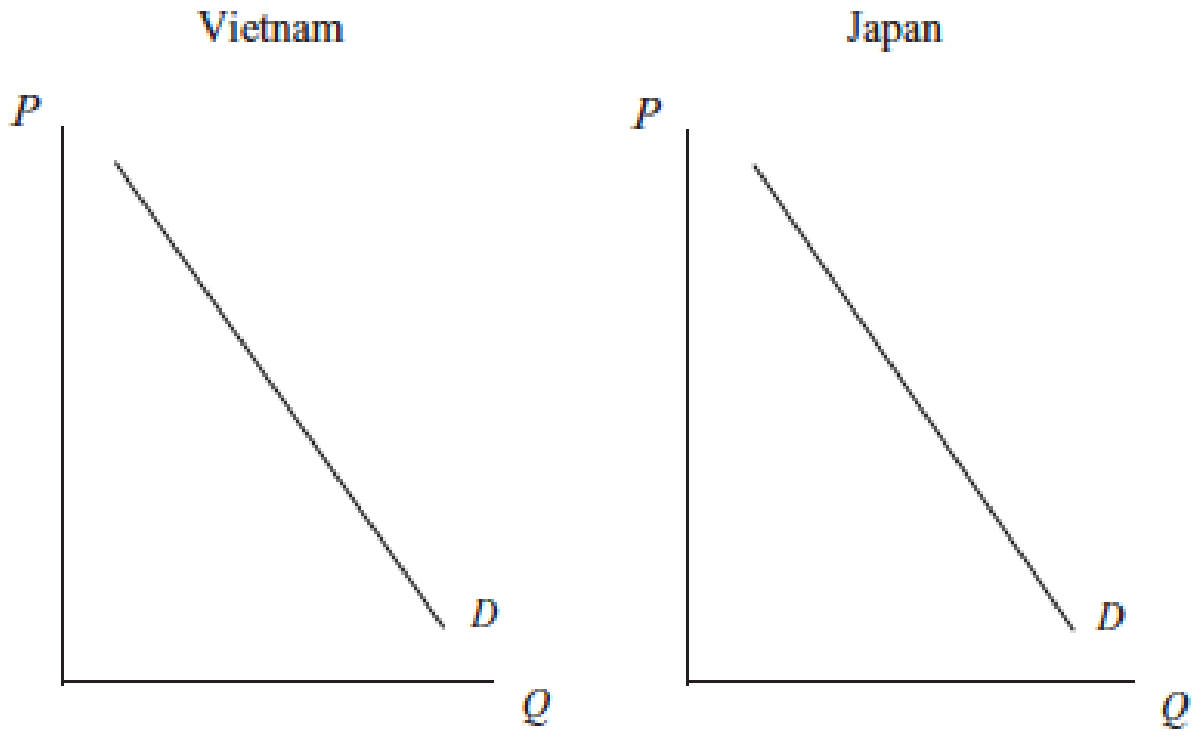

Figure 2.2. Demand for Rice in Vietnam and Japan

Assumptions of the Smith's model:

Difference in production costs (Wage, production knowledge) → Absolute Price difference ← Differences in demand

- The only factor of production is labor (no capital or land). Production capability is fixed (static)
- Taste is the same in two countries (composition of their consumption basket is the same).
- **Perfect competition (so constant return to scale)** **Why?**

Assumptions of the Smith's model:

Difference in production costs (Wage, production knowledge) → Absolute Price difference ← Differences in demand

- The only factor of production is labor (no capital or land). Production capability is fixed (static)
- Taste is the same in two countries (composition of their consumption basket is the same).
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- **No transportation cost**
- **No transaction cost**
- **No government interference in prices**

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- The only factor of production is labor (no capital or land). Production capability is fixed (static)
- Taste is the same in two countries (composition of their consumption basket is the same).
- Perfect competition (so constant return to scale)
- No transportation cost
- No transaction cost
- No government interference in prices
- ***Factor mobility (labor can effortlessly go to the exporting sector).***

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Production chain for an iPhone

Product design

Raw Materials → Intermediate Components (computer chip) → Assembly (Computer chip + other components) → Software, Marketing & retailing →

Production chain for an iPhone

Raw Materials

export

Production chain for an iPhone

Raw Materials → export → Intermediate Components (computer chip)

Production chain for an iPhone

Product design

Raw Materials → export → Intermediate Components (computer chip) → export → Assembly (Computer chip + other components) → export →

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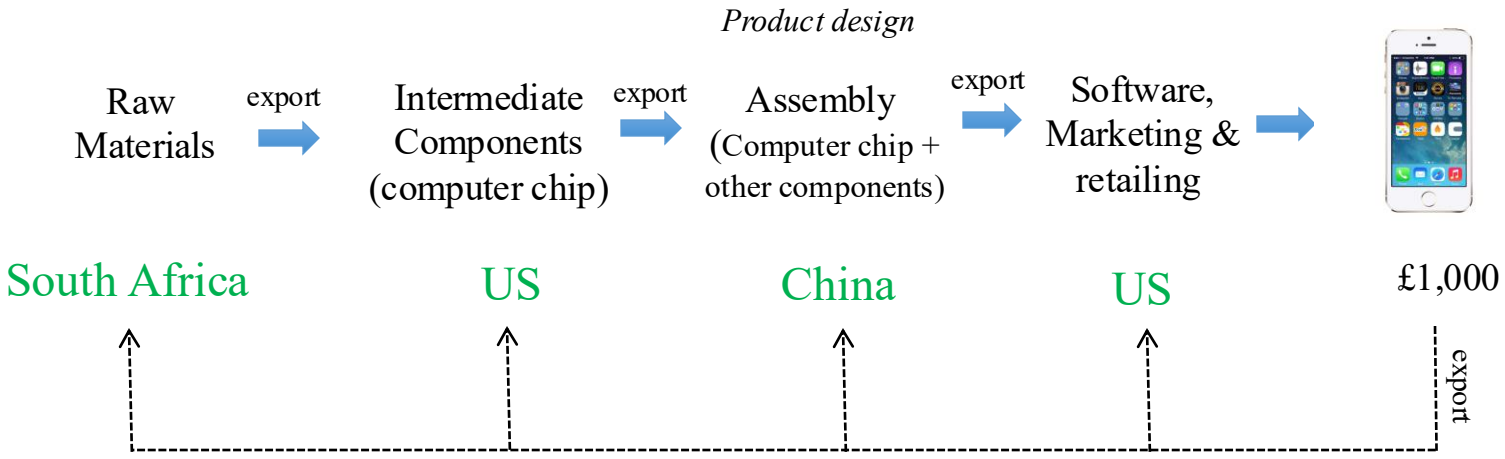
Production chain for an iPhone

Product design

Raw Materials	export →	Intermediate Components (computer chip)	export →	Assembly (Computer chip + other components)	export →	Software, Marketing & retailing	→	
South Africa		US		China		US		£1,000

Production chain for an iPhone



All countries are better-off

(South Africa & China lowers the cost of production for Apple and in return import a chapter and better iPhone than "home iPhone")

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Trump considers 10% tariff on China from February

GETTY IMAGES

João da Silva
Business reporter

22 January 2025

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Who are the the winners and losers of this policy inside the the USA, according to absolute advantage theory?

Why does this policy reduce total welfare in the USA, according to the absolute advantage theory?

Trump considers 10% tariff on China from February

GETTY IMAGES

João da Silva
Business reporter

22 January 2025

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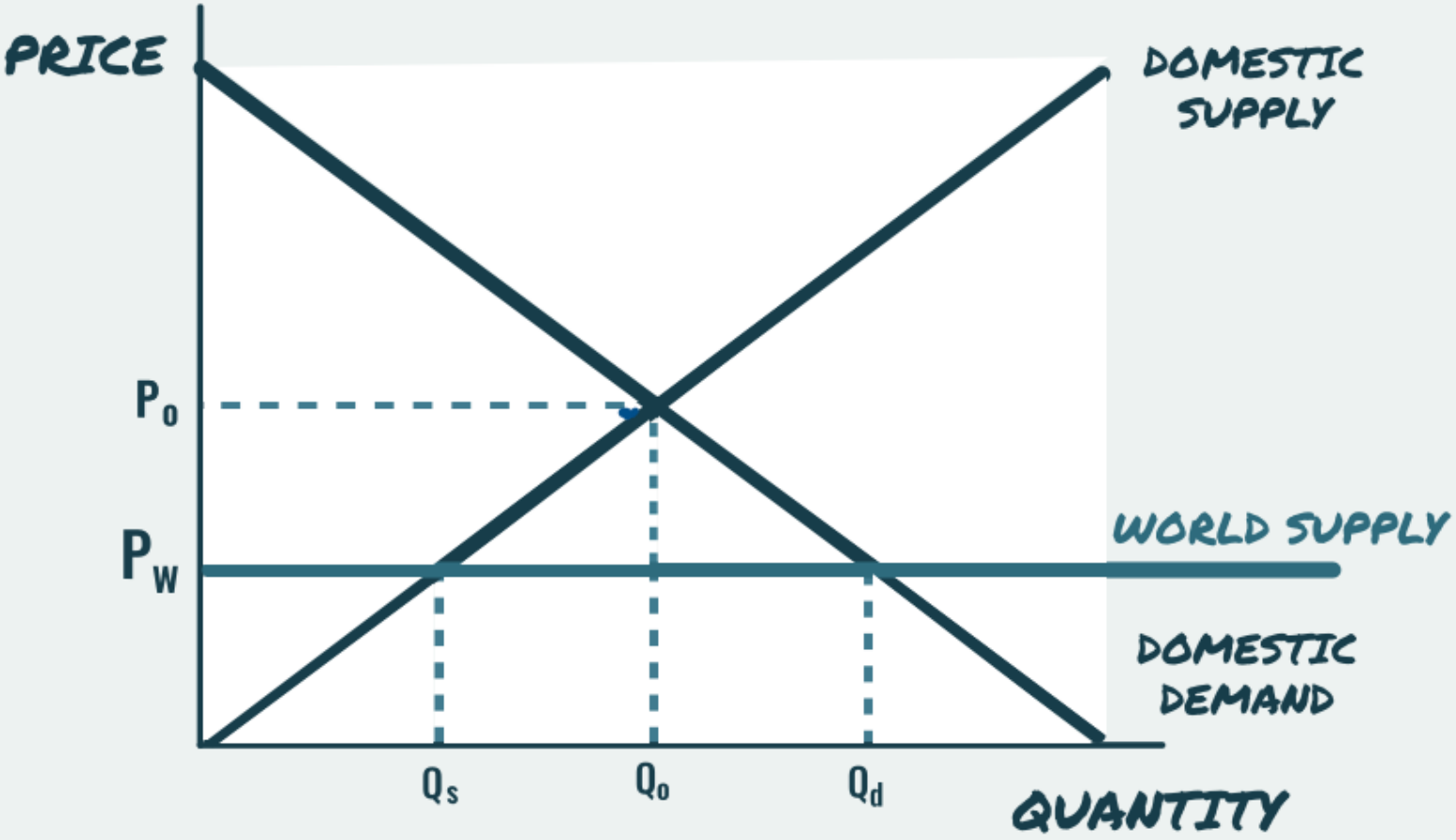
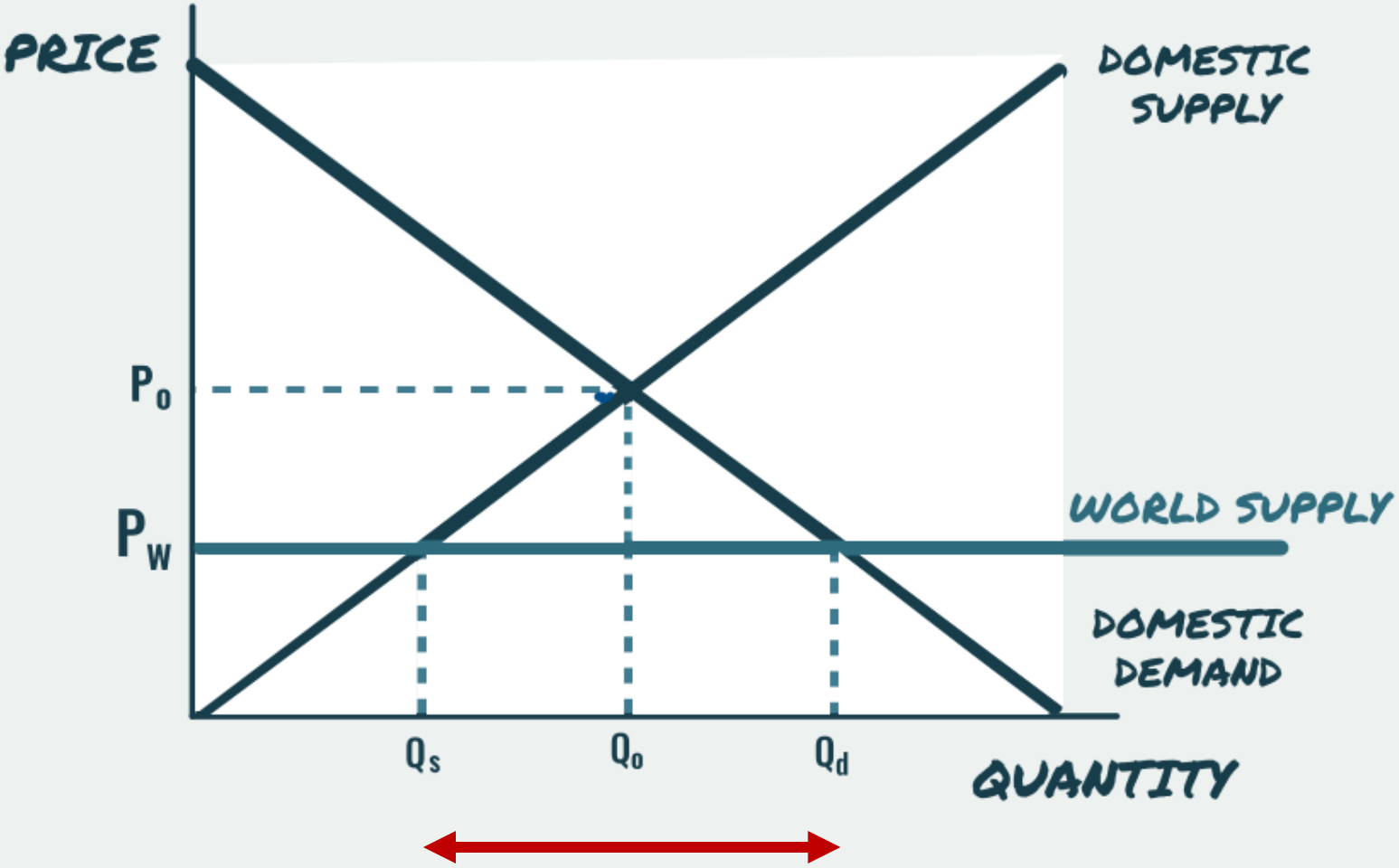
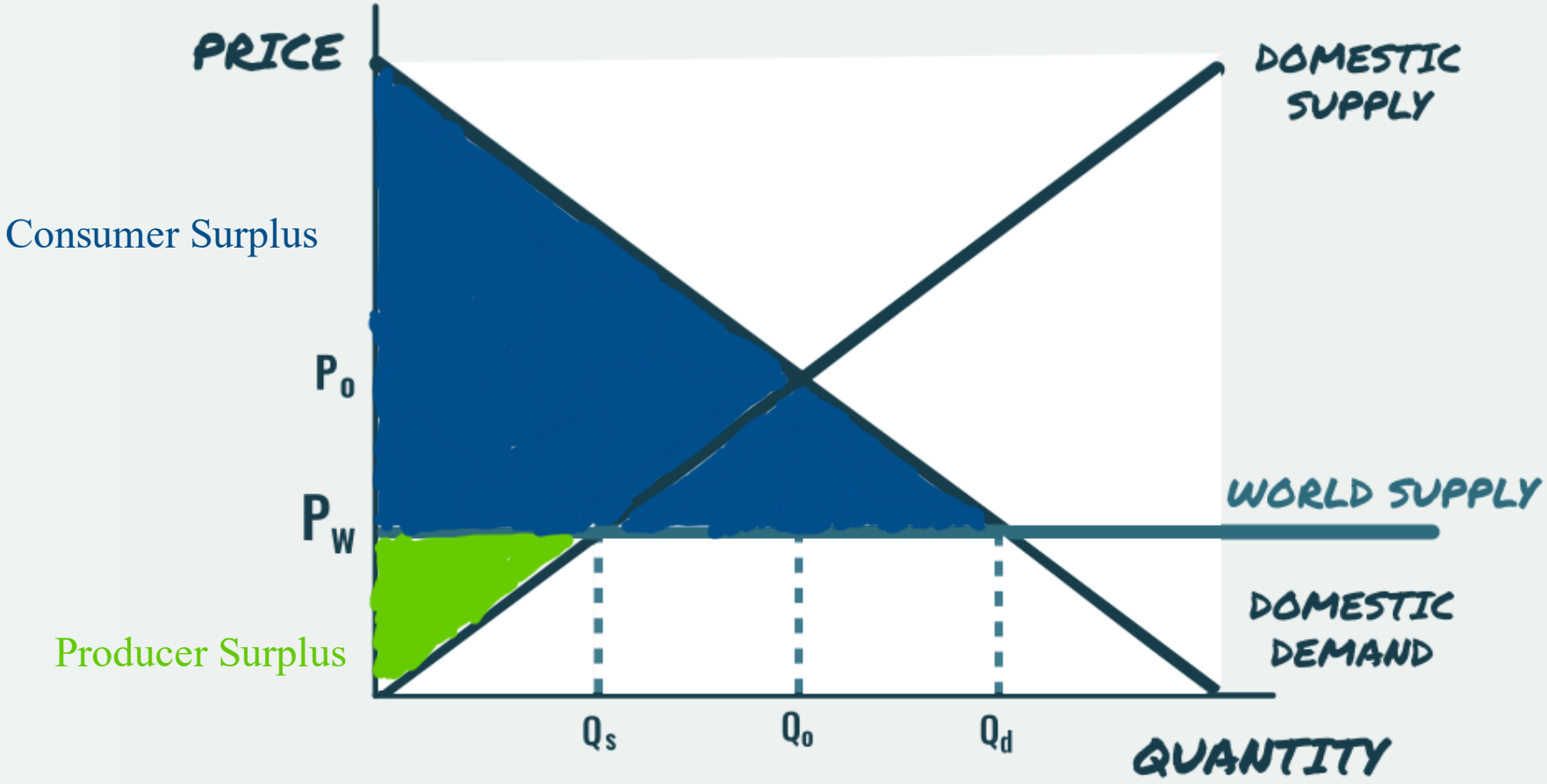
Steel market in the USA, with no tariffs


PRICE
DOMESTIC SUPPLY
P_0
P_W
WORLD SUPPLY
DOMESTIC DEMAND
Q_s
Q_0
Q_d
QUANTITY

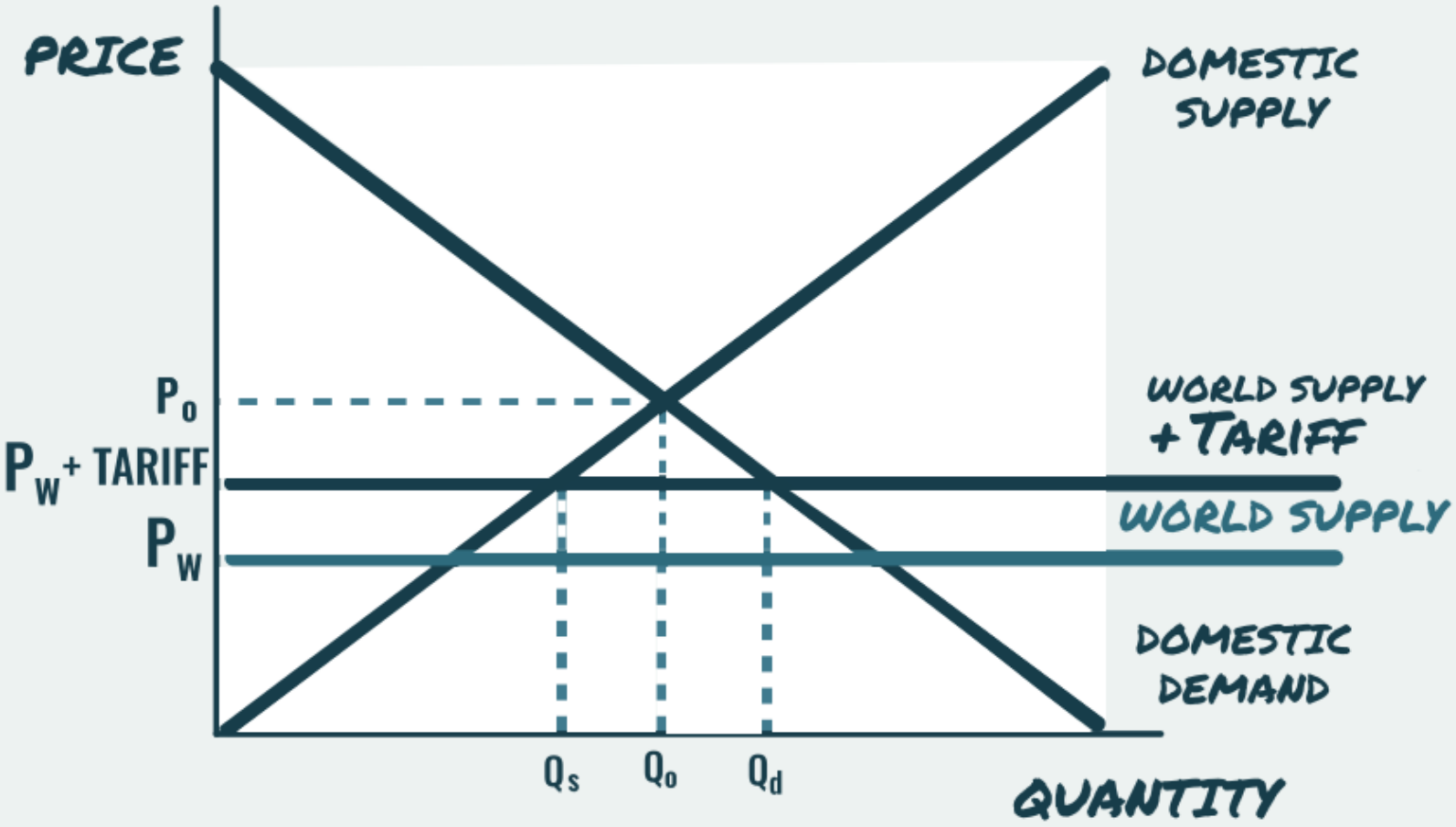


Import of steel to the USA

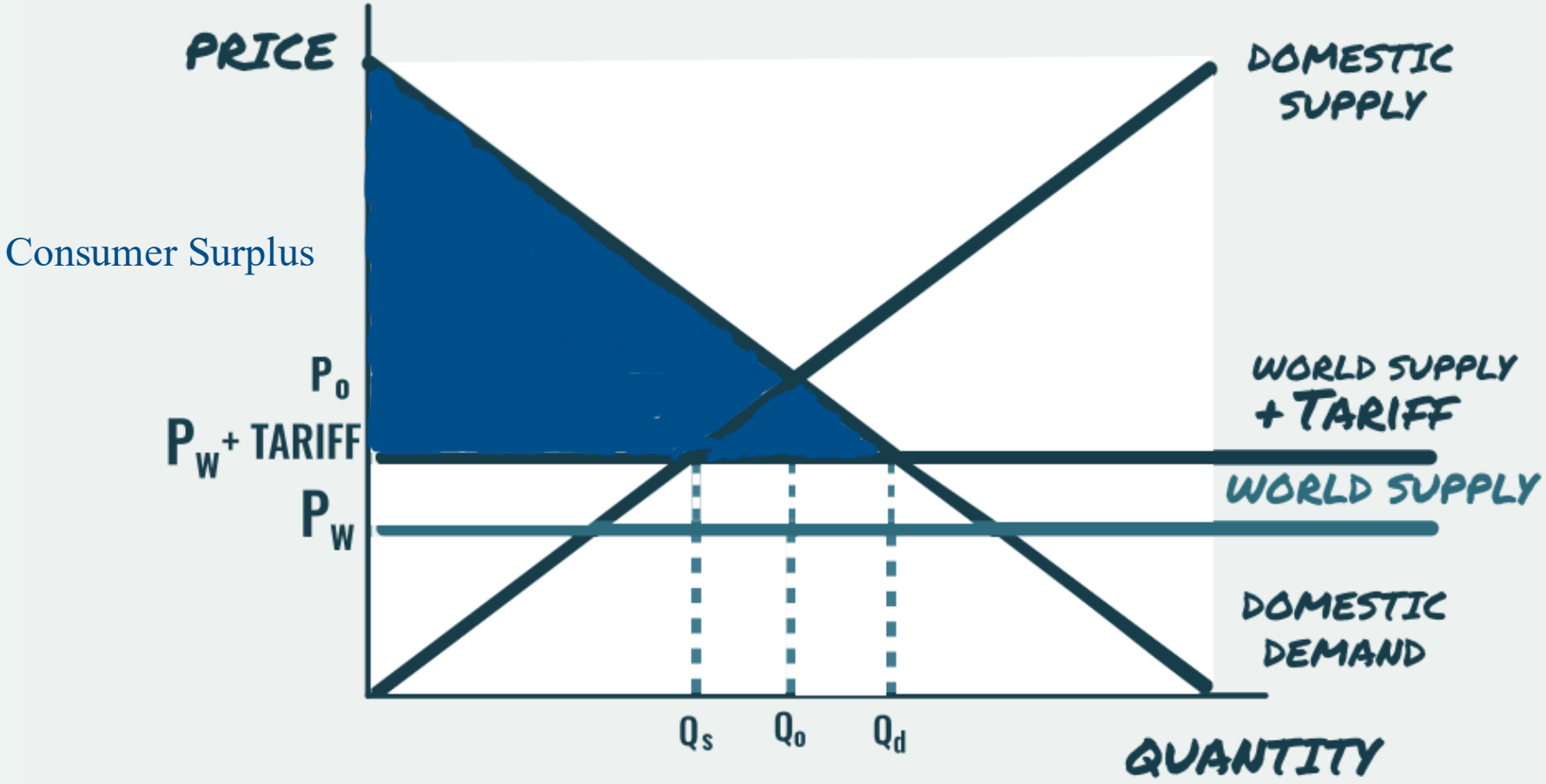

PRICE
DOMESTIC SUPPLY
Consumer Surplus
P_0
P_W
WORLD SUPPLY
DOMESTIC DEMAND
Producer Surplus
Q_s
Q_0
Q_d
QUANTITY

President Trump decides to increase tariff on steel imports

Price of American steel



Price of American steel



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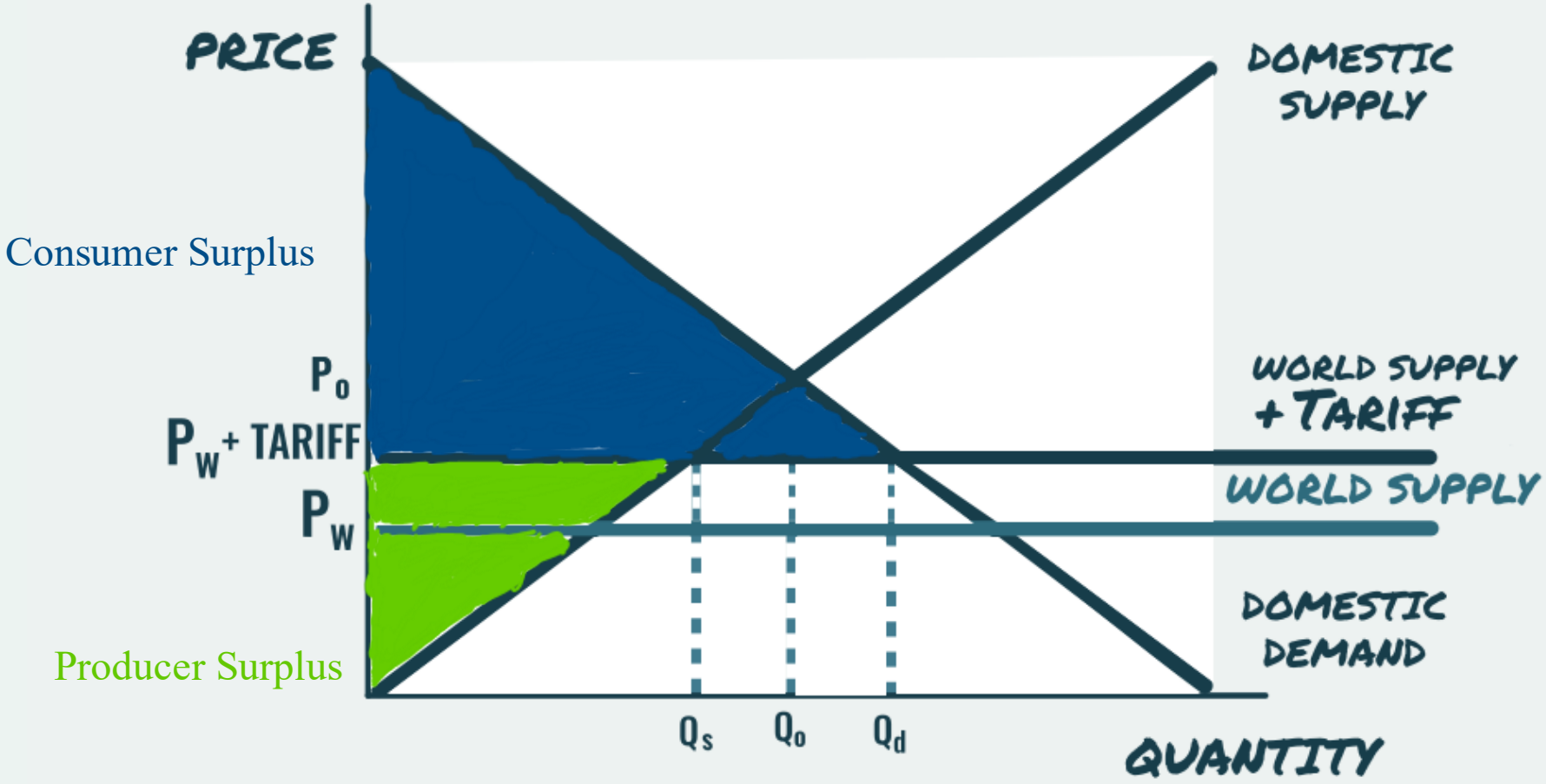
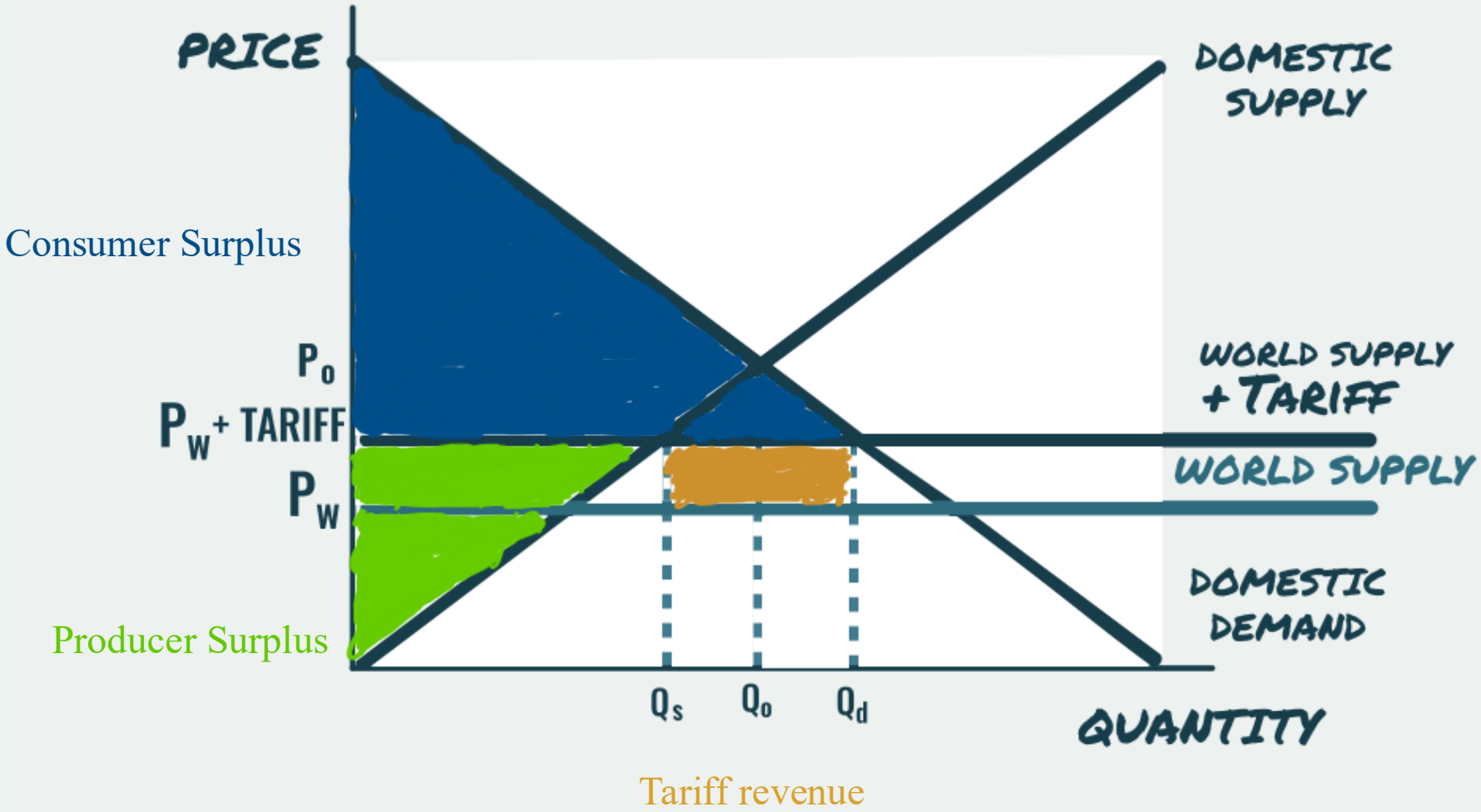
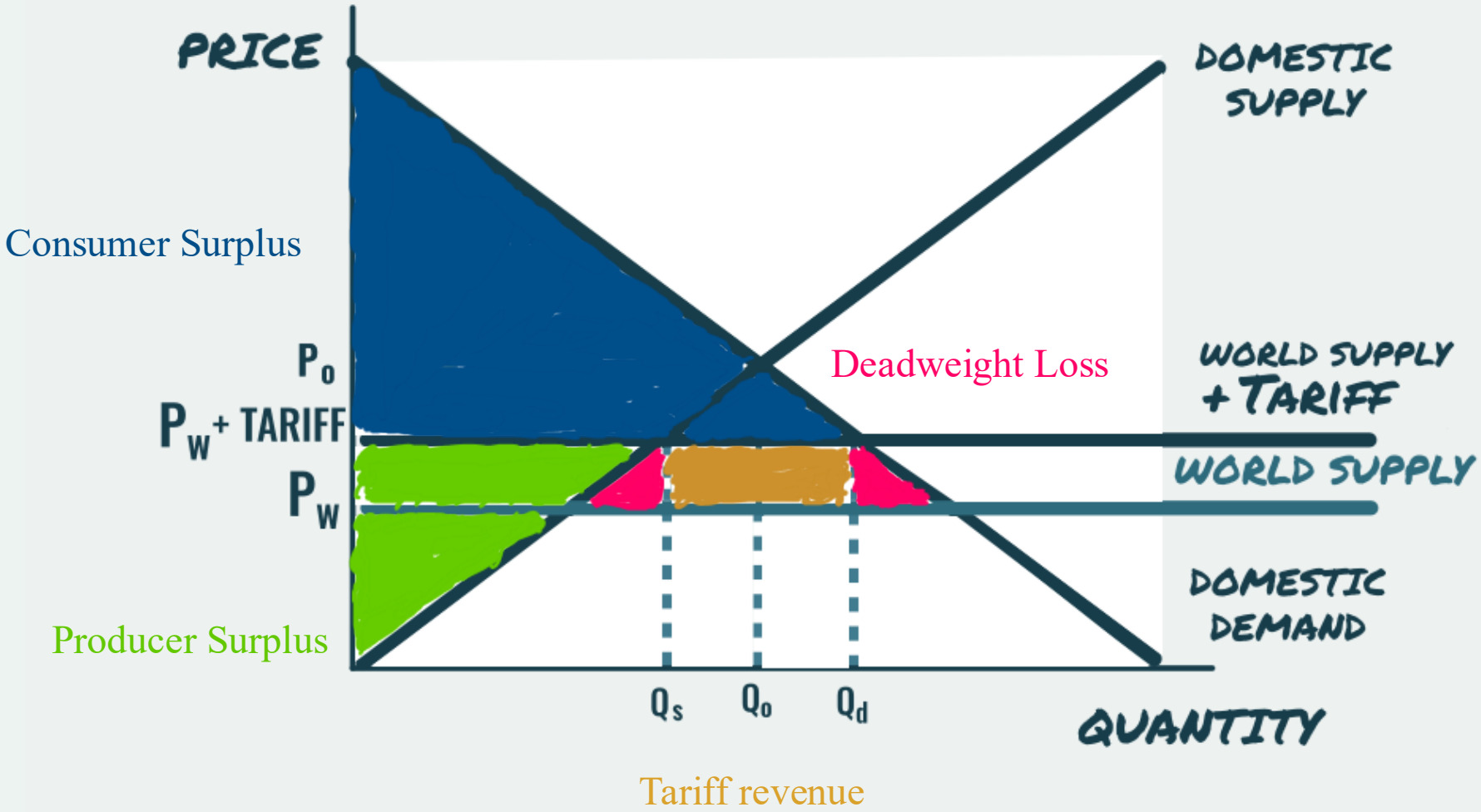



PRICE
DOMESTIC SUPPLY
Consumer Surplus
P_0
P_W+ TARIFF
WORLD SUPPLY + TARIFF
P_W
WORLD SUPPLY
DOMESTIC DEMAND
Producer Surplus
Q_s
Q_0
Q_d
QUANTITY
Tariff revenue


PRICE
DOMESTIC SUPPLY
Consumer Surplus
Deadweight Loss
P_0
P_W+ TARIFF
WORLD SUPPLY + TARIFF
P_W
WORLD SUPPLY
DOMESTIC DEMAND
Producer Surplus
Q_s
Q_0
Q_d
QUANTITY
Tariff revenue

Lecture plan:

- Types of trade
- Theory of absolute advantage
- **Theory of comparative advantage**
- Shortcomings of these theories

Skills Lab 3:
Developing your research idea

Skills Lab 3: *"Developing your research idea"*

Convenor: Dr Ramin Nassehi

Date: Wednesday 4th February,
Time: 2:30 pm to 4:00 pm
Room: TBC
Register:

Why attend?

The ability to conduct independent research is a vital skill for employability. Whether in the corporate sector, public policy, academia, or economic journalism, professional economists rely on strong research skills to excel in their roles. This skill lab will equip you with the foundational tools to embark on your research journey, ensuring you are prepared for challenges and opportunities ahead.

What to expect?

This workshop will guide you through the early stages of your research process:

- Formulating a compelling research question
- Planning around data availability
- Avoiding common pitfalls students face

Register here

Who should attend?
This workshop is ideal for:

- Students aspiring to research careers in the corporate or public sector
- Those interested in economic journalism
- Participants or future contributors to the Explore Econ conference
- Students aiming to publish in *The Economic Tribune* or the *UCL Journal of Economics* (UJE)

Take the first step toward becoming a skilled, independent researcher. Join us! Successful completion of this workshop requires attendance and the submission of a brief post-work assignment, which will be discussed during the session.

Best

Ramin

Register here

	Japan	Vietnam
	$100	$200
	$5	$8

	Japan	Vietnam
	$100	$200
	$5	$8

Relative prices in Japan: 1 motorcycle= 20 kilos of rice (100/5)

Relative prices in Vietnam: 1 motorcycle=25 kilos of rice (200/8)

	Japan	Vietnam
	$100	$200
	$5	$8

Motorcycle is relatively cheaper in Japan.

Relative prices in Japan: 1 motorcycle= 20 kilos of rice (100/5)

Relative prices in Vietnam: 1 motorcycle=25 kilos of rice (200/8)

Rice is relatively cheaper in Vietnam.

	Japan	Vietnam
	$100	$200
	$5	$8

Productivity differences causing relative price differences

Japan is 2 times more productive than Vietnam in producing motorcycle ($200/$100)

Japan is 1.6 time more productive than Vietnam in producing rice ($8/$5)

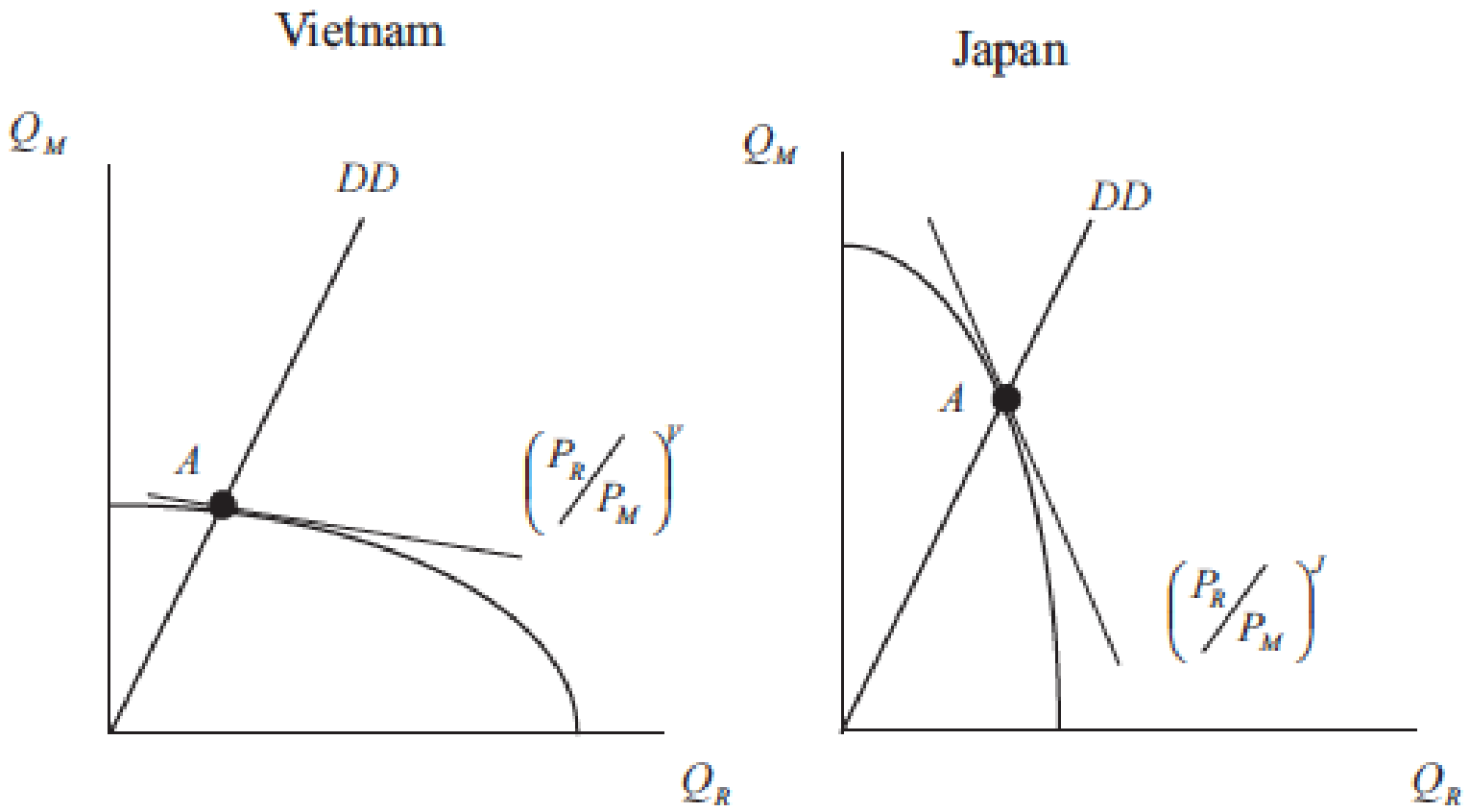


Figure 3.3. Relative Prices in Vietnam and Japan under Autarky

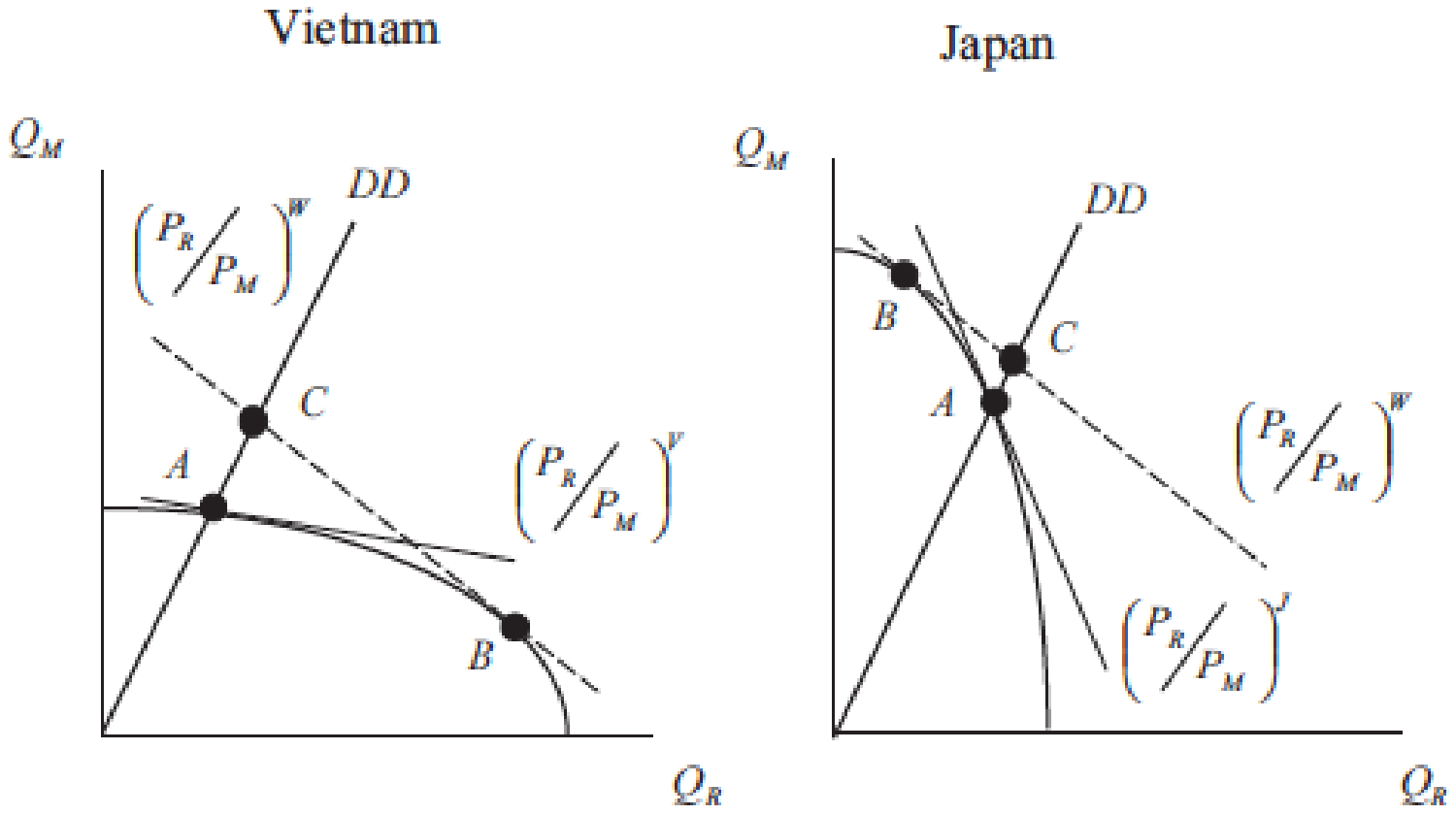


Figure 3.4. Autarky and Comparative Advantage in Vietnam and Japan

By exporting, each country can import more goods than it could have produced locally.

Theory of comparative advantage

David Ricardo
(1772-1823)

Even when one country can produce *everything* cheaper than another country, it can still benefit by specializing in production of goods or tasks in which its has the *greatest productivity/cost advantage* and import goods (or tasks) in which it has less productivity advantage.

Theory of comparative advantage

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Even when one country can produce *everything* cheaper than another country, it can still benefit by specializing in production of goods or tasks in which its has the *greatest productivity/cost advantage* and import goods (or tasks) in which it has less productivity advantage. **Countries export goods at a higher *relative* price so that they can import goods at a lower *relative* price.** Specialisation and trade enables countries to *consume* more at the aggregate level!

Theory of comparative advantage

David Ricardo
(1772-1823)

Do what you do best and import the rest!!

UCL Economics

TO HONOUR THE MEMORY OF
DAVID RICARDO
1772 - 1823
UCL DEPARTMENT OF
ECONOMICS
FOUNDED 1828
Unveiled during the Econometric Society World Congress 2005

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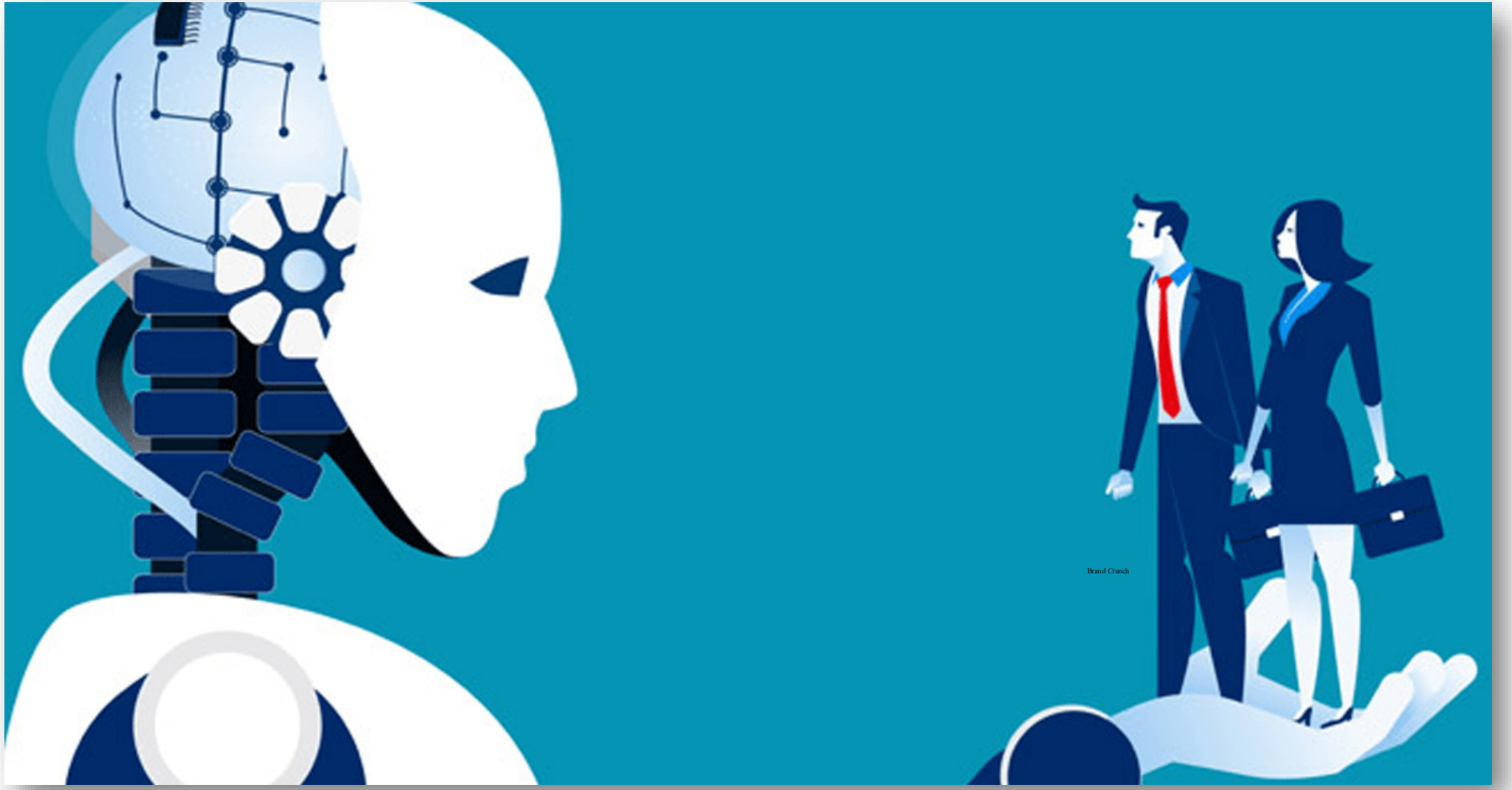
- No transportation cost
- No transaction cost
- No government interference in prices
- Factor mobility inside the country (but not between countries)

Milton Friedman - Imports, Exports & Exchange Rates

CLICK HERE

[redacted]

The upcoming AI revolution leading to reversal of outsourcing?



Production chain for an iPhone

Product design

Raw Materials → Intermediate Components (computer chip) → Assembly (Computer chip + other components) → Software, Marketing & retailing →

Production chain for an iPhone

Product design

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Q: Apple is considering spending $1 billion to build robots in US that can assemble iPhones cheaper than India. Should Apple:

1. Build these robots and assemble products inside USA
2. Don’t build these robots and outsource assembly to India

Join at menti.com | use code 5106 4692

Q: Apple is considering spending $1 billion to build robots in US that can assemble iPhones cheaper than India. Should Apple:

0	0
Build these robots and assemble products inside USA	Don't build these robots and outsource assembly to India

Responding is off

Press C to allow new responses

CLICK HERE

End presentation

Production chain for an iPhone

Product design

Raw Materials → Intermediate Components (computer chip) → Assembly (Computer chip + other components) → Software, Marketing & retailing →

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These slides are left blank intentionally…..

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Lecture plan:

- Types of trade
- Theory of absolute advantage
- Theory of comparative advantage
- **Shortcomings of these theories**

Shortcoming I: Trade liberalization has no social pain. There is only one factor of production, which effortlessly move across sectors.

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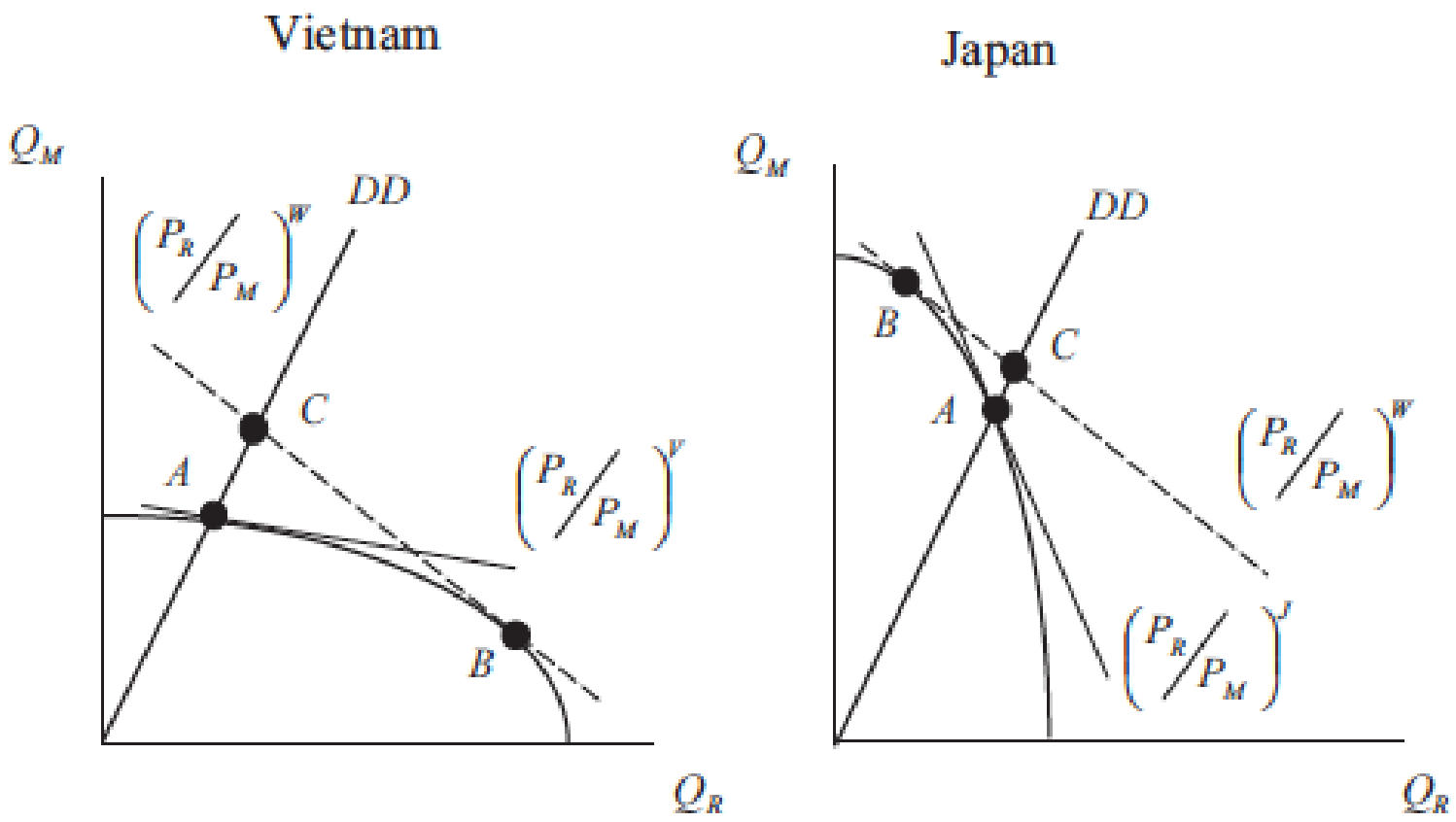

Figure 3.4. Autarky and Comparative Advantage in Vietnam and Japan

Shortcoming II: absolute advantage theory and comparative advantage theory are static. They don’t tell us what happens over time.

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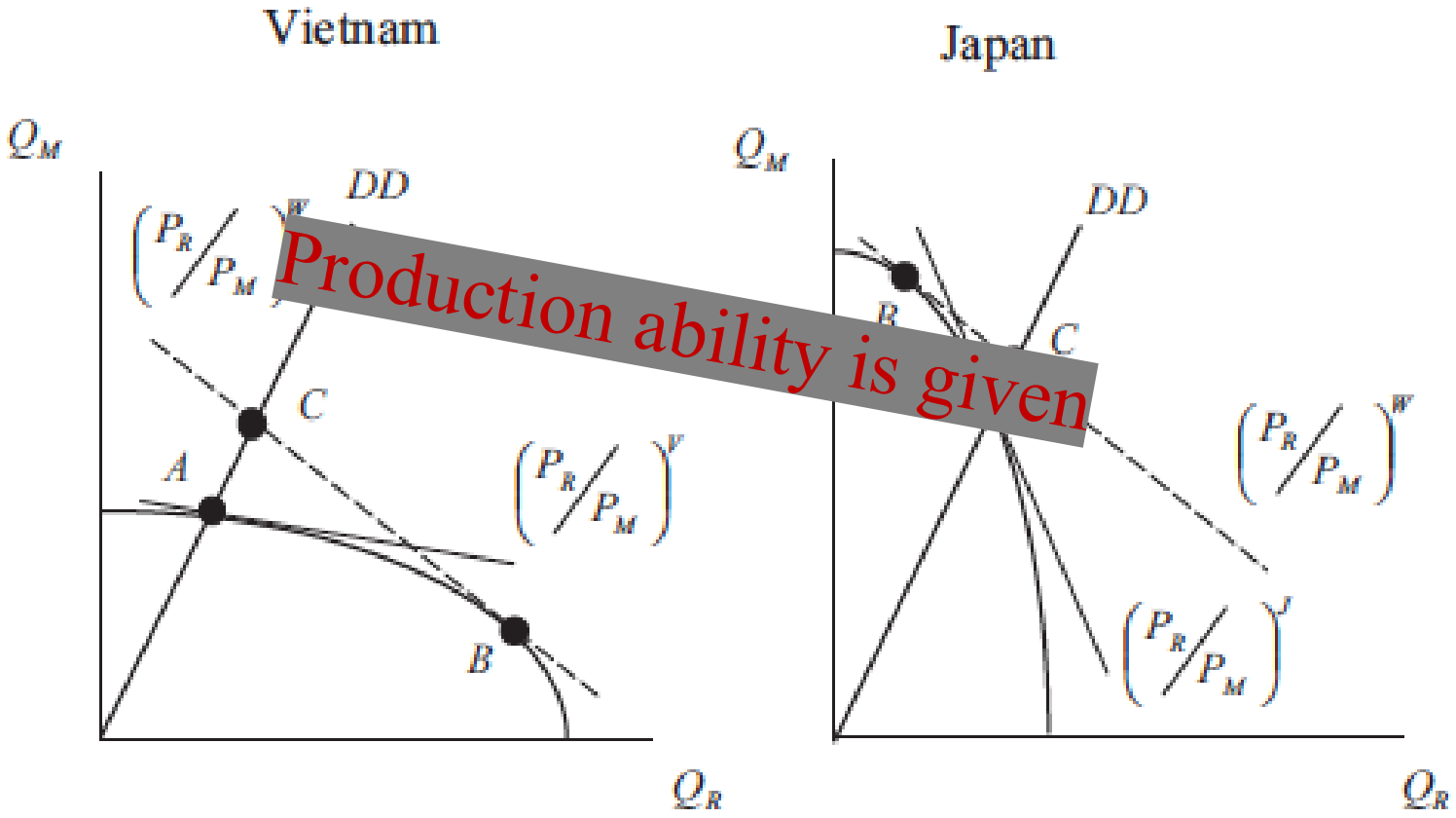

Figure 3.4. Autarky and Comparative Advantage in Vietnam and Japan

Summary

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- Shortcoming II: They don’t say much about about long-term term impact of trade.

Appendix

The Impact of the 2018 Tariffs on Prices and Welfare

Mary Amiti

Stephen J. Redding

David E. Weinstein

JOURNAL OF ECONOMIC PERSPECTIVES
VOL. 33, NO. 4, FALL 2019
(pp. 187–210)

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Abstract

We examine conventional approaches to evaluating the economic impact of protectionist trade policies. We illustrate these conventional approaches by applying them to the tariffs introduced by the Trump administration during 2018. In the wake of this increase in trade protection, the United States experienced substantial increases in the prices of intermediates and final goods, dramatic changes to its supply-chain network, reductions in availability of imported varieties, and the complete pass-through of the tariffs into domestic prices of imported goods. Therefore, the full incidence of the tariffs has fallen on domestic consumers and importers so far, and our estimates imply a reduction in aggregate US real income of $1.4 billion per month by the end of 2018. We see similar patterns for foreign countries that have retaliated with their own tariffs against the United States, which suggests that the trade war has also reduced the real income of these other countries.

The Impact of the 2018 Tariffs on Prices and Welfare

Mary Amiti

Stephen J. Redding

We find that by December 2018, import tariffs were costing US consumers and the firms that import foreign goods an additional $3.2 billion per month (£10 per citizen per month) in added tax costs and another $1.4 billion per month in deadweight welfare (efficiency) losses ($4 per citizen per month)

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The Impact of the 2018 Tariffs on Prices and Welfare

Mary Amiti

Stephen J. Redding

We estimate that the combined effects of input and output tariffs have raised the average price of US manufacturing by 1 percentage point, which compares with an annual average rate of producer price inflation from 1990 to 2018 of just over 2 percentage points.

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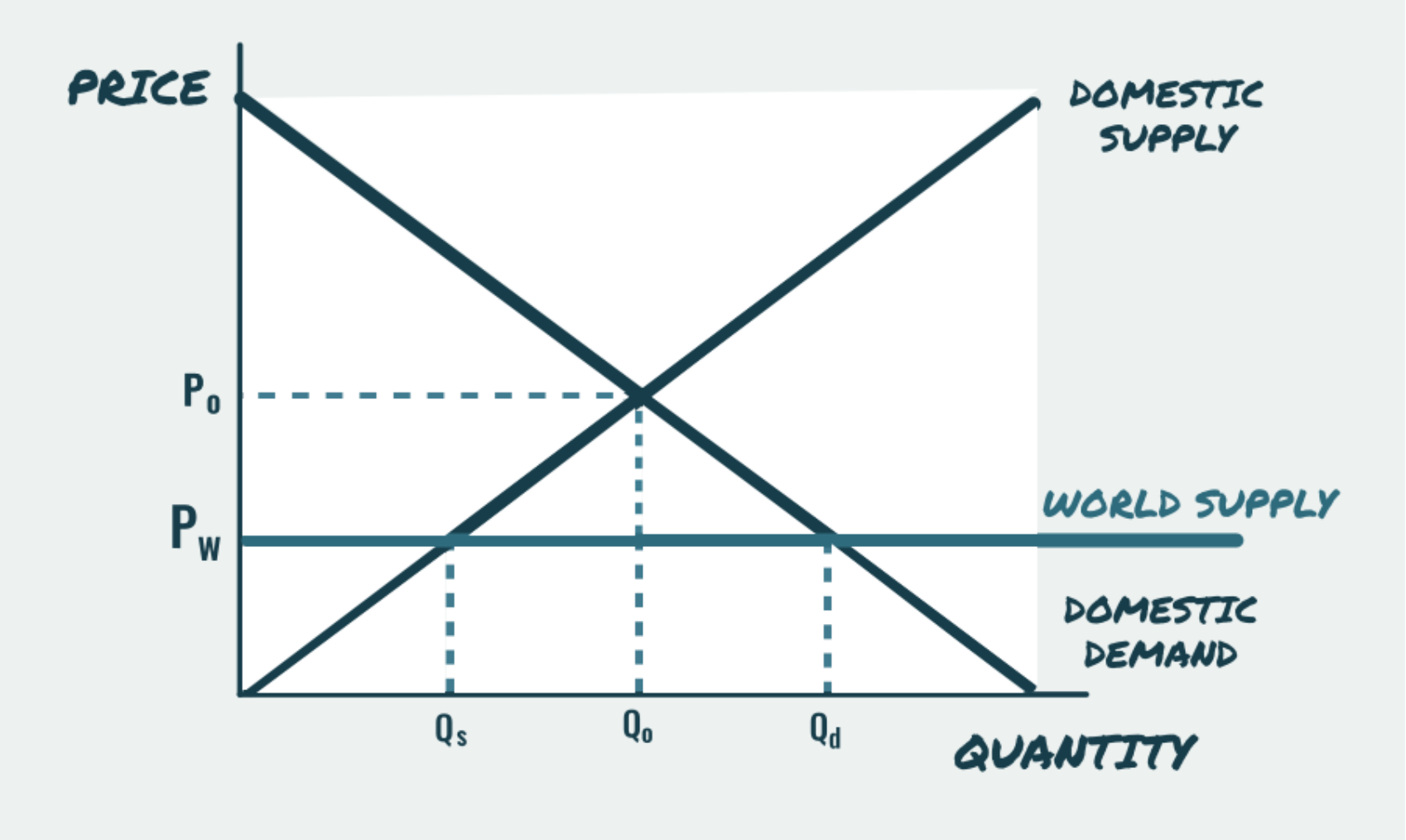
Steel market in the USA, when there is no trade (autarky)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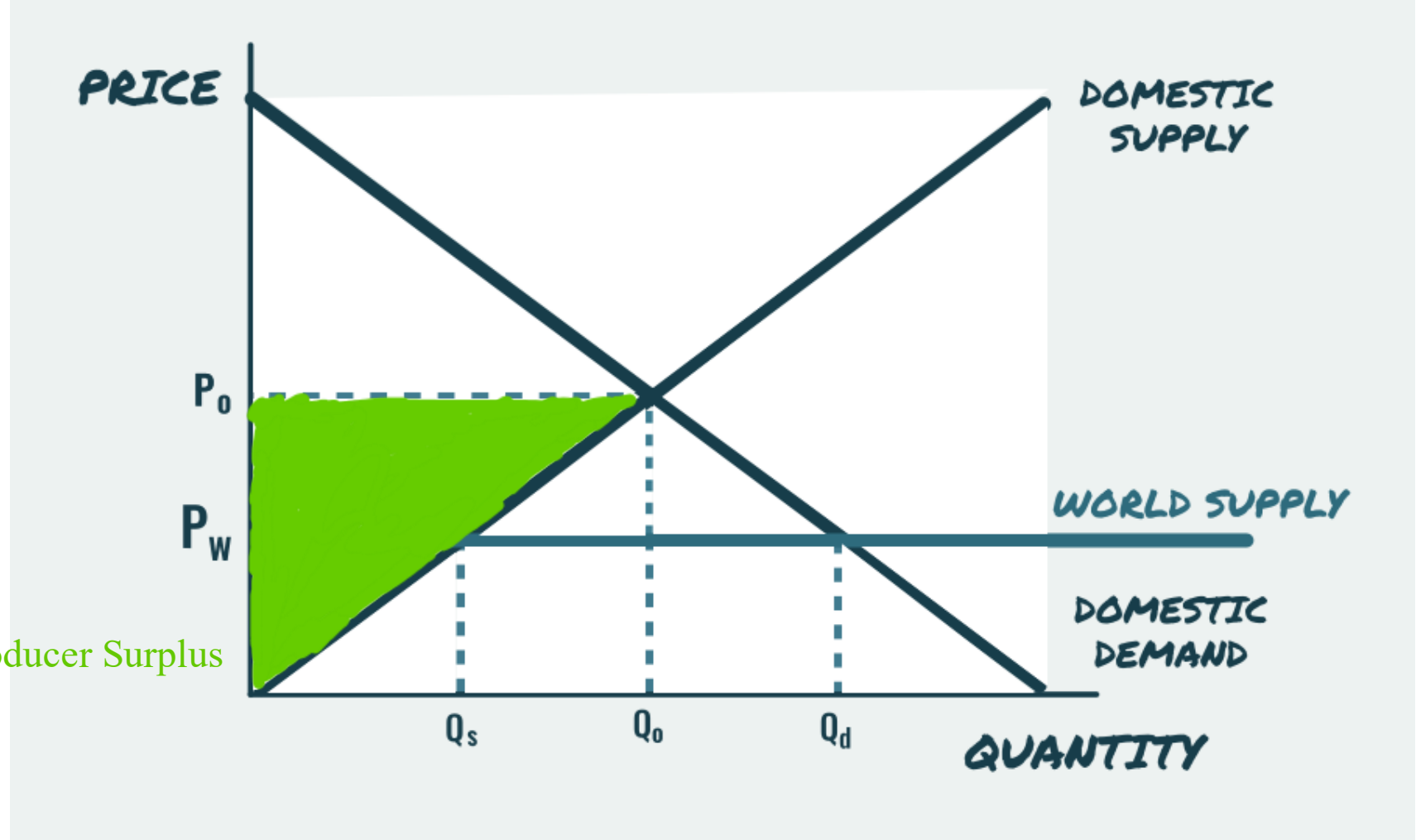
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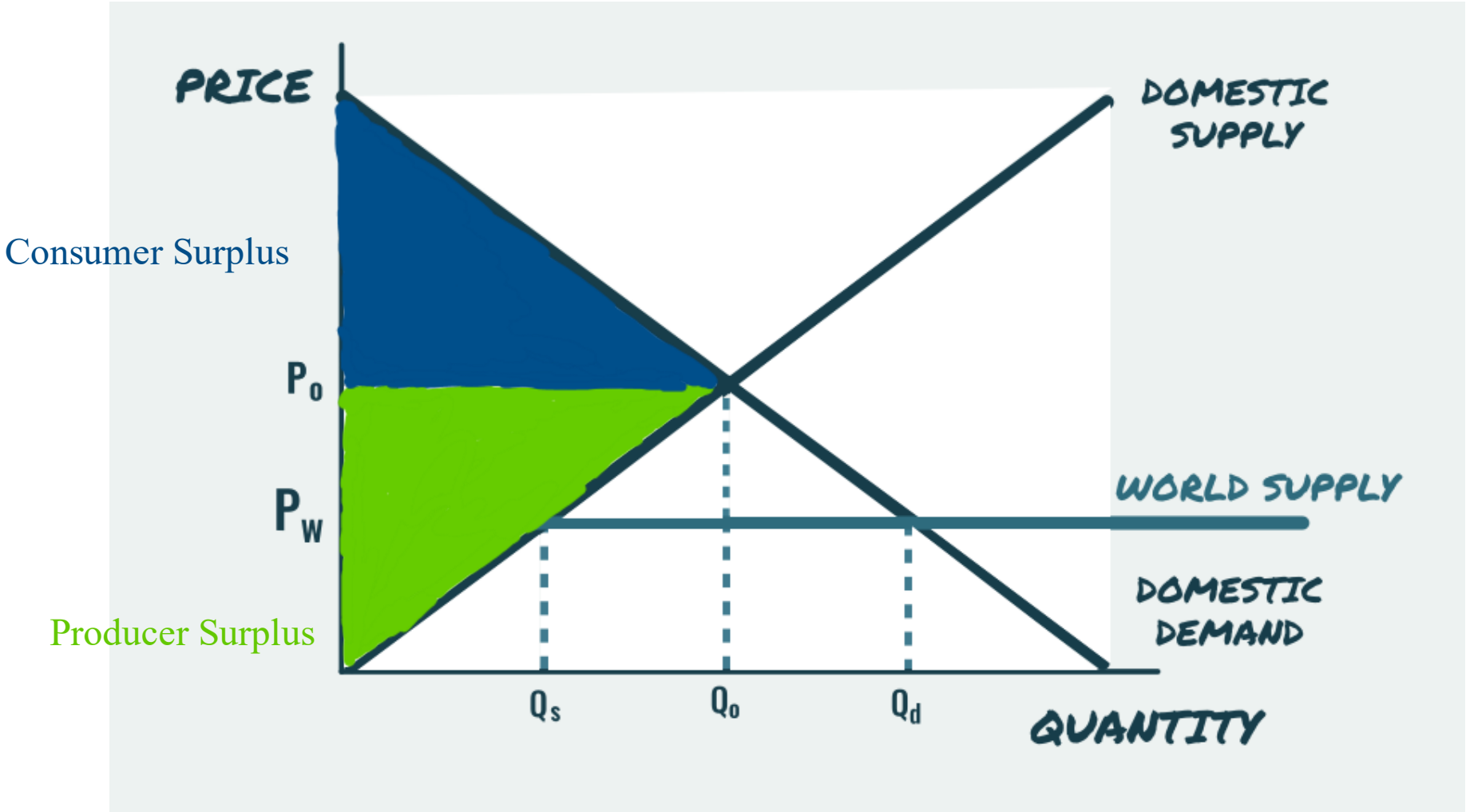
Consumer Surplus
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